



CSE 598 008 EECS 498 003 Lab 4

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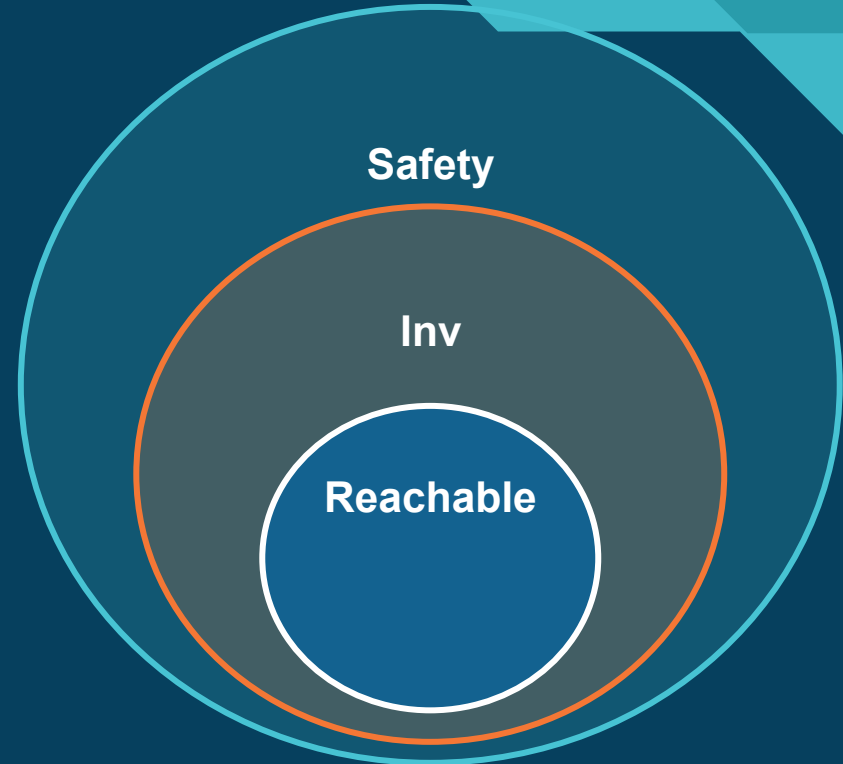
Based on slides by Ivan Wei (Fall 2025).

Reminder

- Problem Set 1 due tonight, September 25, 11:59pm
- Problem Set 2 released September 26, due October 7, 11:59pm
- Midterm will be on October 22, 6pm – 8pm, FMCRB 1050
- Last-minute office hours for PS1 after the exercises

Inductive Invariant

- Goal: Safety holds in every reachable state
- First try: induction on Safety itself
 - $\text{Init}(v) \implies \text{Safety}(v)$
 - $\text{Safety}(v) \ \&\& \ \text{Next}(v, v') \implies \text{Safety}(v')$
- Problem: Safety alone is often **not** inductive: a safe state may step to an unsafe one
- Fix: find a stronger invariant Inv:
 - $\text{Init}(v) \implies \text{Inv}(v)$
 - $\text{Inv}(v) \ \&\& \ \text{Next}(v, v') \implies \text{Inv}(v')$
 - $\text{Inv}(v) \implies \text{Safety}(v)$



Proof Exercise

- Part A: Crawler
- Part B: Rusty Bridge, continued from Lab 3
- Starter code: `lab04a.dfy` and `lab04b.dfy` on the course GitHub

Tips - Split by Step

```
lemma NextPreservesInv(v: Variables, v': Variables)
  requires Inv(v)
  requires Next(v, v')
  ensures Inv(v')
{
  var step :| NextStep(v, v', step);
  match step
  case Action1Step(<parameters>) => ...
  case Action2Step(<parameters>) => ...
}
```

- This is one reason we use JNF: Next is a single exists over Step, so `:|` can pick the step and match can split on it
- Splitting by step shows which action breaks Inv
- Add assert statements to see what Dafny can and cannot prove
- A recursive function over a sequence may need a helper lemma proved by induction

Tips - Triggers

- Dafny only uses a forall fact when a term matching its trigger appears in the proof

- Example:

```
ghost function f(x: int): int
```

```
lemma Magic()
```

```
  requires forall x :: f(x) > 0
```

```
  ensures exists y :: f(y) > 0
```

```
{
```

```
  assert f(0) > 0;
```

```
}
```

- Without the assert, Dafny fails: no $f(\dots)$ term appears, so the forall is never used
- This is why a single line can be “magical” in finishing a proof, and why Dafny sometimes seems “soooooo stupid”

Extra - MU Puzzle

- Start with the string MI
- Rules (x and y stand for any strings):
 - Rule 1: $xI \rightarrow xIU$ (if the string ends in I, add a U)
 - Rule 2: $Mx \rightarrow Mxx$ (double everything after the M)
 - Rule 3: $xIIIIy \rightarrow xUy$ (replace IIII with U)
 - Rule 4: $xUUy \rightarrow xy$ (delete UU)
- Apply the rules any number of times, in any order
- Example: $MI \rightarrow MII \rightarrow MIIII \rightarrow MUI$
- Question: can you ever reach MU?